Kotha Rishika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.78
Tirumala junior college	Class XII (Board of Intermediate Education, Andhra Pradesh - BIEAP)	2024	98.90%
Tirumala school	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2022	97.00%

PROJECTS

AI Research Assistant (RAG) | *Python - LangChain - FAISS - Sentence Transformers - Streamlit*

- Developed a **Retrieval-Augmented Generation (RAG)** application for research paper analysis, integrating **PDF parsing, semantic indexing, vector retrieval, and LLM-based question answering** using **LangChain, FAISS, Sentence Transformers, OpenAI API, and Google Gemini API**.
- Implemented an **end-to-end document ingestion pipeline** comprising **PDF extraction (PyMuPDF), text chunking, embedding generation, vector database construction, semantic similarity search, and context-aware response generation**.
- Built a **Streamlit-based web application** supporting document ingestion, conversational retrieval, and real-time interaction with research papers.

Customer Churn Prediction Pipeline | *Python - Scikit-learn - MLflow - Airflow - Docker - Streamlit*

- Developed an **end-to-end MLOps pipeline** for telecom customer churn prediction, integrating **data preprocessing, feature engineering, model training, experiment tracking, automated retraining, and interactive model serving** using **Scikit-learn, MLflow, Apache Airflow, Docker, and Streamlit**.
- Containerized the application using **Docker** and managed project dependencies with **Poetry**, ensuring environment reproducibility and deployment portability.
- Built a **Streamlit-based inference application** with **interactive Plotly dashboards**, enabling real-time churn prediction and model performance visualization.

Image Classification -- Kaggle Competition (**Group Project)**

- Built a **CNN from scratch in PyTorch** using depthwise-separable convolutions, Batch Normalization, SiLU activations, Dropout2d, and Adaptive Average Pooling for binary image classification.
- Developed a reproducible training pipeline with **data augmentation, stratified validation, AdamW, weight decay, learning-rate scheduling, and gradient clipping** for improved generalization and training stability.
- Collaborated in a **3-member team** on architecture and hyperparameter experimentation, achieving **21st place out of 74 teams** on the IIT Hyderabad Deep Learning Kaggle leaderboard.

Deep Learning Architectures from Scratch | *Python, NumPy, PyTorch*

- Implemented **CNN, Vanilla RNN, LSTM, Transformer Encoder, Linear SSM, and Selective SSM** from scratch using **Python, NumPy, and PyTorch**, including custom convolution, BPTT, multi-head self-attention, positional encoding, and sequence modeling.
- Conducted **optimizer (SGD, Momentum, RMSProp, Adam), regularization (BatchNorm, Dropout), and architecture ablation studies**, analyzing optimization, gradient flow, and long-context sequence modeling.
- Achieved **74.1% CNN training accuracy**, **Transformer F1-score of 0.667**, **LSTM F1-score of 0.647**, and **53% Selective SSM accuracy** on image and sentiment classification benchmarks.

SKILLS

Programming Languages: Python, C, SQL, HTML

Frameworks/Libraries: Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch, Seaborn, Langchain

Tools: Git, GitHub, Docker, Google Colab, MySQL, LaTeX

Operating Systems: Windows, Linux

Coding Platform: You can view my Leetcode profile here

RELEVANT COURSES

Artificial Intelligence: Intro to Modern AI, Classical AI, Programming for AI, Foundation of Machine Learning, AI for Humanity, Deep Learning

Computer Science: Introduction to Programming, DBMS 1, OS-1, Data Structure and Algorithms

Mathematics: Discrete Mathematics, Calculus-1, Calculus-2, Vector Calculus, Series of Functions, Probability and random variables, Differential equations, Matrix Theory, Intro to Metric spaces, Numerical Analysis

Others: Linear Systems and Signal Processing, Control Systems

POSITIONS OF RESPONSIBILITY

Sunshine - Buddy

- Coordinated club activities and events while collaborating with team members to ensure smooth planning and execution.
- Demonstrated teamwork, time management, and problem-solving skills while fulfilling organizational responsibilities.

EXTRACURRICULAR

Actively participated in community service initiatives, including **Swachh Bharat** drives, walkathons, and orphanage visits, promoting social responsibility and community engagement.